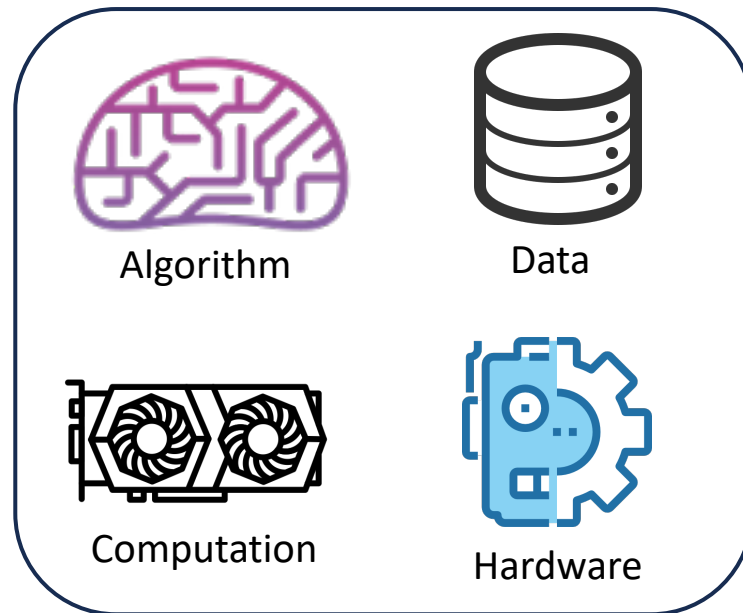


Robot Learning: What, Why, and How?



Embodied intelligence in the physical world



Instructor: Guanya Shi

Recording Reminder

- ☐ This session will be audio recorded for educational use by other students in this course.

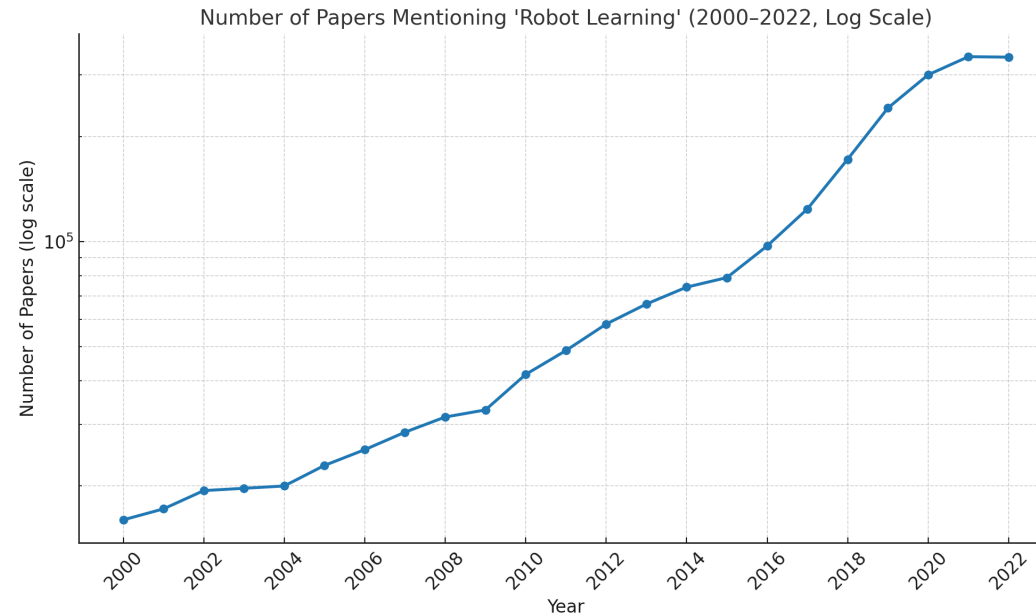
Welcome to the First Day!

- ❑ For how many of you, it is the first semester at CMU?



Logistics

- ❑ The largest 16-831 so far, since we started the course in Fall 2022
 - 160 students enrolled, 76 WL!!!
 - Before Fall 2022 it was called “16-831: Statistical Techniques in Robotics”
- ❑ We will offer the same class in Spring 2027
 - Please consider taking the spring one!
 - If you plan to drop, don't wait until deadline 🙏
- ❑ Feel free to audit!



Logistics

- ❑ Slides are available before each lecture on Piazza or website
- ❑ No Canvas. Only (1) Website, (2) Piazza, and (3) Gradescope

- ❑ Website: <https://16-831.github.io/>
- ❑ Piazza (only for officially registered students): <https://piazza.com/cmu/fall2026/16831>
 - Email TAs if you haven't been enrolled yet!
- ❑ Gradescope (only for officially registered students):
<https://www.gradescope.com/courses/1353766>
 - Email TAs if you haven't been enrolled yet!

- ❑ Recording (Panopto): Will update the link on Piazza soon

Logistics

- ❑ This course will be recorded but only for students
- ❑ Please try your best to attend lectures in person
 - Maximize your learning
 - There will be a lot of blackboard derivations/illustrations
- ❑ Some slides are from Deepak Pathak (F23), David Held (F22), and Katerina Fragkiadaki (CMU 10-703), and other courses
- ❑ Speak up and participate! Raise your hands for any questions!

Take Care of Yourself!

- ☐ It is natural to struggle with stress and mental health challenges.
 - ☐ All of us benefit from support during times of struggle.
 - ☐ You are not alone.
 - ☐ Asking for support sooner rather than later is often helpful.
 - ☐ Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit their website at <http://www.cmu.edu/counseling/>.
-
- ☐ Feel free to chat about life, career plan, research ideas, and more!

Course Organizations

- ❑ All the details will be on the website
 - Office hour (OH)
 - Latest course schedule and due dates
 - Optional reading (please do read them!)
 - Late policy (5 late days, but ONLY for four HW assignments), collaboration policy (AI is viewed as a collaborator in this course), academic integrity, etc.

Collaboration Policy

Collaboration is encouraged, but the work you submit for assignments is expected to be entirely your own. The writing and code must be yours, and you must fully understand everything that you submit. Discussing a paper or how to solve a problem is fine, but you must write your submission yourself. List everyone you collaborated with in the assignment write-up. Highly identical work without proper attribution will be handled according to university policy. See the [CMU Academic Integrity Guidelines](#).

Generative AI tools such as ChatGPT are viewed as collaborators in this course and should be acknowledged accordingly.

- ❑ Assessment:
 - 60% Four assignments (15% each)
 - 20% Project presentation (last two lectures)
 - 20% Project final report

Course Structure

Latest schedule: See the course website

- ☐ Topic group 1 (2 lectures): Course introduction: What is robot learning and why?
- ☐ Group 2 (2 lectures): Machine/Deep learning refresher
- ☐ Group 3 (2 lectures): Imitation learning
- ☐ Group 4 (6 lectures): Model-free reinforcement learning
- ☐ Group 5 (4 lectures): Model-based reinforcement learning
- ☐ Group 6 (2 lectures): Bandit, preference-based learning, exploration
- ☐ Group 7 (2 lectures): RL from offline data
- ☐ Group 8 (4 lectures): Specialized topics (two guest lectures)
- ☐ Project presentations

Course Project

- ❑ Team size: 4-6 people (team size ≤ 3 is only allowed under exceptional circumstances).
- ❑ Two tracks:
 - Reproduce an existing robot learning paper and implement some improvement ideas on top of it (the ideas may or may not work)
 - Open-ended project
- ❑ Three components:
 - 1-2 page project proposal (due on 10/13): It will carry NO grade, but is mandatory to submit for early feedback.
 - Project oral presentation (12/01 and 12/03)
 - 6-8 page project final report (due on 12/11, in CoRL template)
- ❑ CMU SCS might be able to offer some AWS credits (<\$150 per student). We will make a Google form soon to collect information.

Guanya Shi

- ❑ Ph.D. from Caltech (Control and Dynamical Systems)
- ❑ Assistant Professor @ CMU RI; Amazon Scholar @ Frontier AI & Robotics
- ❑ Research area: learning and control for agile robotics
 - Theoretical and algorithmic foundations
 - Adaptive, reliable, and efficient algorithms
 - Real-world general-purpose robots with agility (e.g., humanoids)
- ❑ Email: guanyas@andrew.cmu.edu
- ❑ OH: Via email appointment. Office: NSH 3209



TA Team



Annabella Macaluso



Esme Rubinstein



Jingtian Yan



Pranay Gupta



Arthur Fender Coelho Bucker



Jimmy Dai

Esme Rubinstein

Hello!

- ❑ 2nd year PhD student
- ❑ Advised by Stephen Smith
- ❑ Work on multi-agent scheduling and coordination
- ❑ Fun facts: This summer, I did a 4 day trek in Georgia (the country) in the mountains
- ❑ E-mail: erubinst@andrew.cmu.edu
- ❑ OH: TBA



Jingtian Yan

Hello!

- ❑ 3rd year PhD student
- ❑ Advised by Jiaoyang Li and Stephen Smith
- ❑ Work on planning and multi-agent system
- ❑ Fun facts: I learned to row under the coaching of a world champion
- ❑ E-mail: jingtiay@andrew.cmu.edu
- ❑ OH: TBA



Jimmy Dai

Hello!

- ❑ 2nd year PhD student
- ❑ Advised by Prof. Guanya Shi (LeCAR)
- ❑ Work on dexterous manipulation
- ❑ Fun facts: I make amazing dumplings
- ❑ E-mail: cunxid@andrew.cmu.edu
- ❑ OH: TBA



Pranay Gupta

Hello!

- ❑ 3rd year PhD student
- ❑ Advised by Prof. Henny Admoni (HARP)
- ❑ Work on *Test-time policy adaptation through human-interaction*
- ❑ Fun facts: My best time on NYT mini is 23 sec
- ❑ E-mail: pranaygu@andrew.cmu.edu
- ❑ OH: Tuesday 5-5:45PM, NSH 4511



Annabella Macaluso

Hello!

- ❑ 2nd year PhD Student
- ❑ Advised by Prof. Yonatan Bisk
- ❑ Work on generalizable learning for robotics.
- ❑ Fun facts: I enjoy long distance biking! 🚴
- ❑ E-mail: amacalus@andrew.cmu.edu
- ❑ OH: TBA



Arthur Bucker

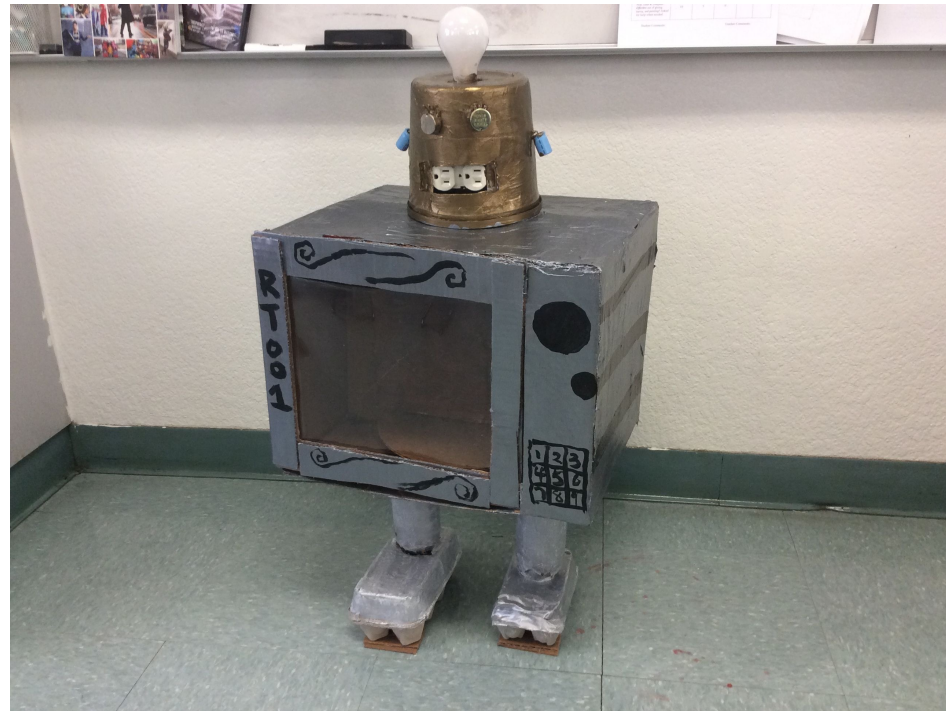
Hello!

- ❑ 4th year PhD student
- ❑ Advised by Prof. Jean Oh (BIG)
- ❑ Work on *Test-time adaptation and Policy Improvement for Manipulation*
- ❑ Fun facts: For my first paper, I had to backflip and parkour while being chased by drones.
- ❑ E-mail: abucker@andrew.cmu.edu
- ❑ OH: Mondays 3:15-4pm, RI-SQH, Room 103



Quick Questions For You

- ☐ Join a live poll at [menti.com](https://www.menti.com)!
- ☐ A very important warm-up question: Is microwave a robot?



Lecture 1&2's Agenda

- ☐ What is robot learning and what is its uniqueness?
- ☐ What is the goal of robot learning?
- ☐ Where are we today? How far are we from the goal?
- ☐ Course structure and course objective
- ☐ Applications of sequential decision making in other domains

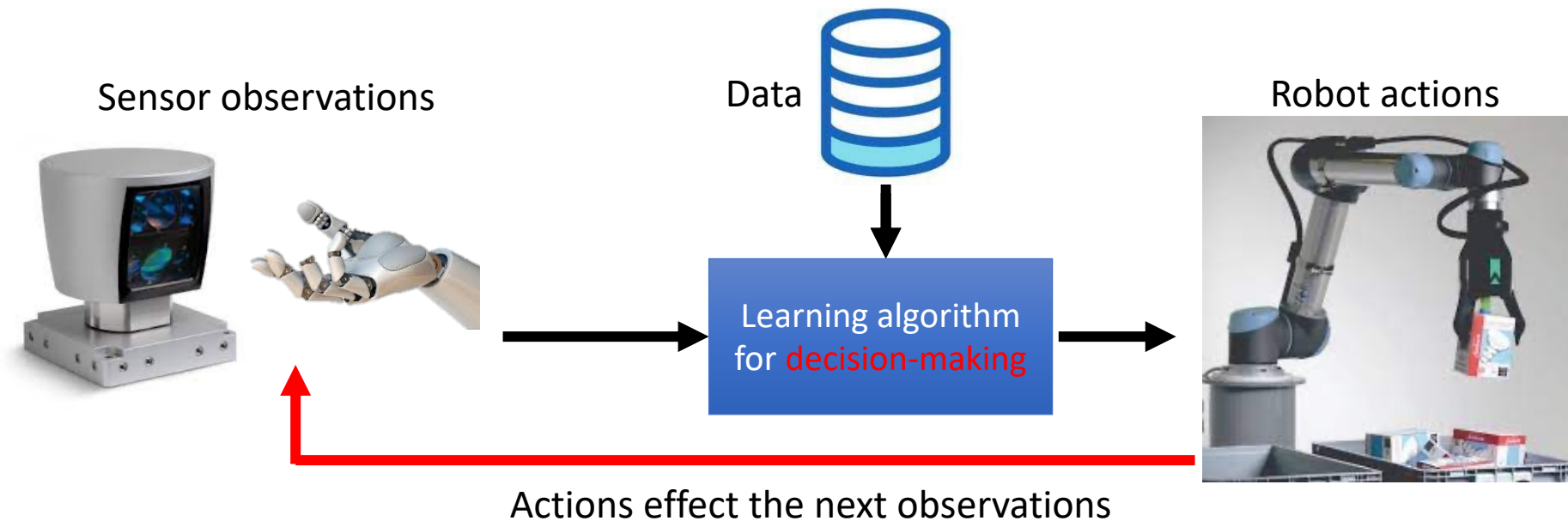
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Let's Start!

What is “robot learning” and what is this class about?

❑ **Learning** to make **sequential** decisions in the **physical world**



What is “Robot Learning”?

- ❑ **Learning** to make **sequential** decisions in the **physical world**
- ❑ **Learning**: Data-driven and improve from data
 - W/o learning & data: search & planning, classic control, ...
- ❑ **Sequential**: The current action/decision influences the next state therefore the next action/decision
 - W/o sequential: bandit, standard supervised learning, ...
- ❑ **Physical world**: The robot needs to interact with the physical world in the closed-loop
 - A.k.a. “embodied intelligence”
 - W/o physical world: RL for games, LLMs...

Uniqueness of “Robot Learning”

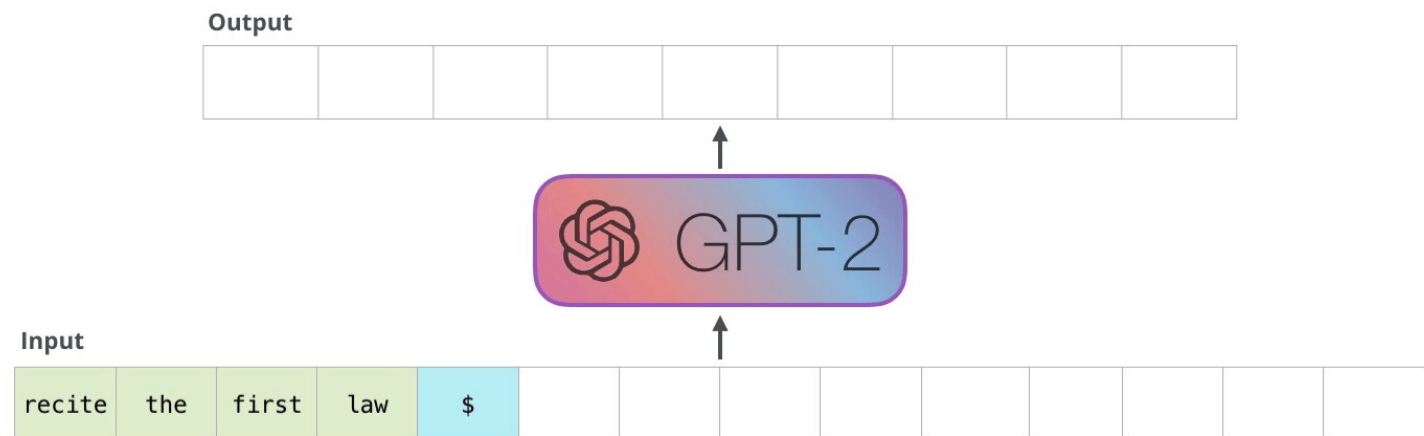
- ❑ **Learning:** Data-driven and improve from data
 - W/o learning & data: search & planning, optimization, classic control, ...

Challenges (compared to other AI or machine learning domains):

- ❑ Where is the data from?
- ❑ How to use the data?

Why do GPTs (and other VLMs/LLMs) work? How much does it apply to robot learning?

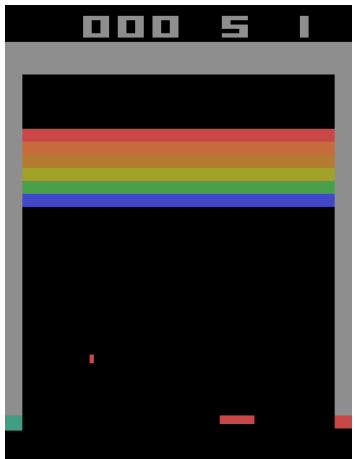
- ❑ Architecture: transformer
- ❑ Data: web texts, books, wikis, ...
- ❑ Loss: next token prediction
- ❑ Optimization: SGD
- ❑ Generation: autoregressive



Uniqueness of “Robot Learning”

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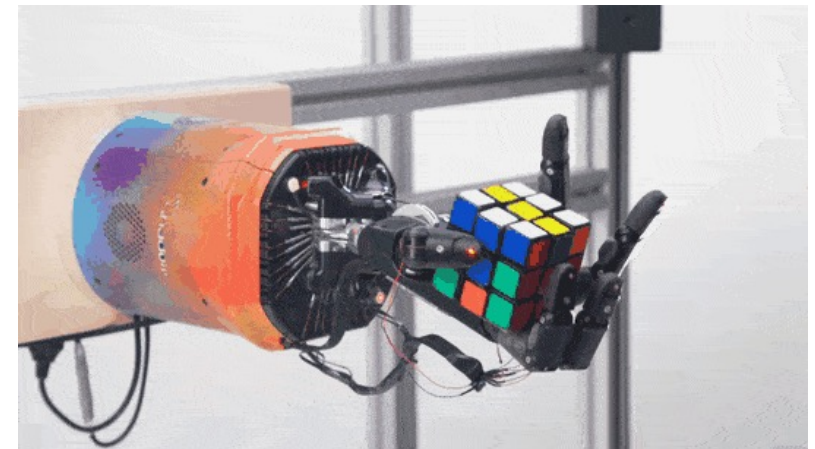
What are challenges (compared to DRL in games such as Alpha Go)?



DQN for Atari (DeepMind)



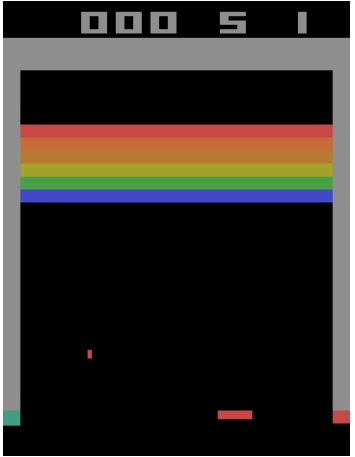
AlphaGo (DeepMind)



Manipulator (OpenAI)

Uniqueness of “Robot Learning”

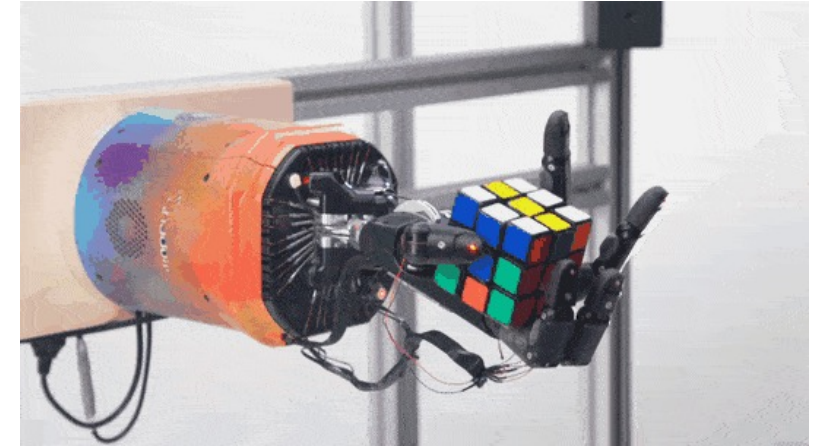
What are challenges (compared to DRL in games such as Alpha Go)?



DQN for Atari (DeepMind)



AlphaGo (DeepMind)



Manipulator (OpenAI)

- ☐ Known & static v.s. Unknown & dynamic
- ☐ One specific task v.s. Many tasks
- ☐ Clear optimization goals (i.e., reward) v.s. Unclear goals
- ☐ Offline learning is enough v.s. Require online adaptation
- ☐ Slow action v.s. Relatively fast real-time action (e.g., 50Hz)
- ☐ Allow failures v.s. Physics doesn't forgive!
- ☐ Discrete digital world v.s. Continuous physical world

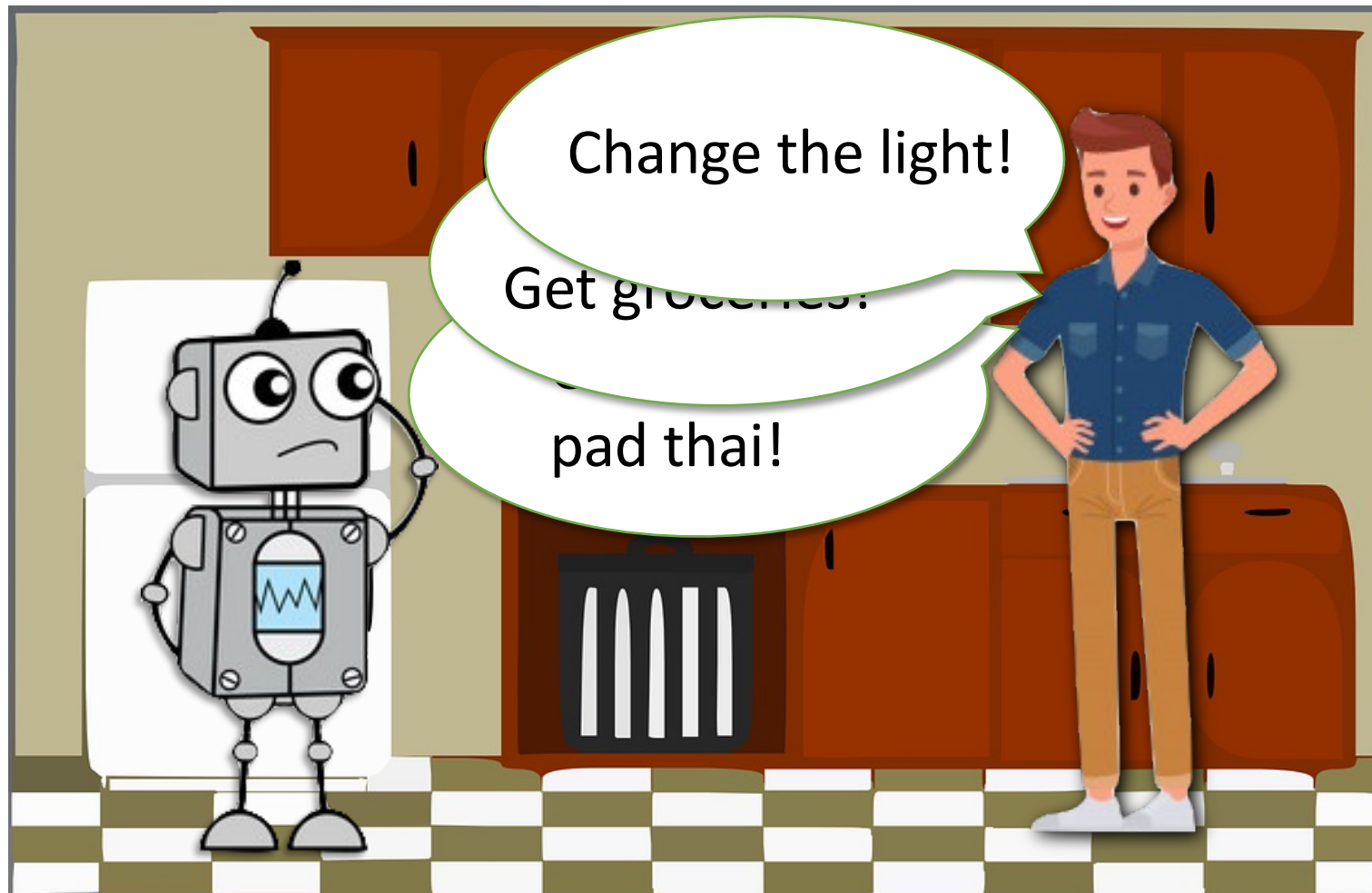


Lecture 1&2's Agenda

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What is the Goal of “Robot Learning”?

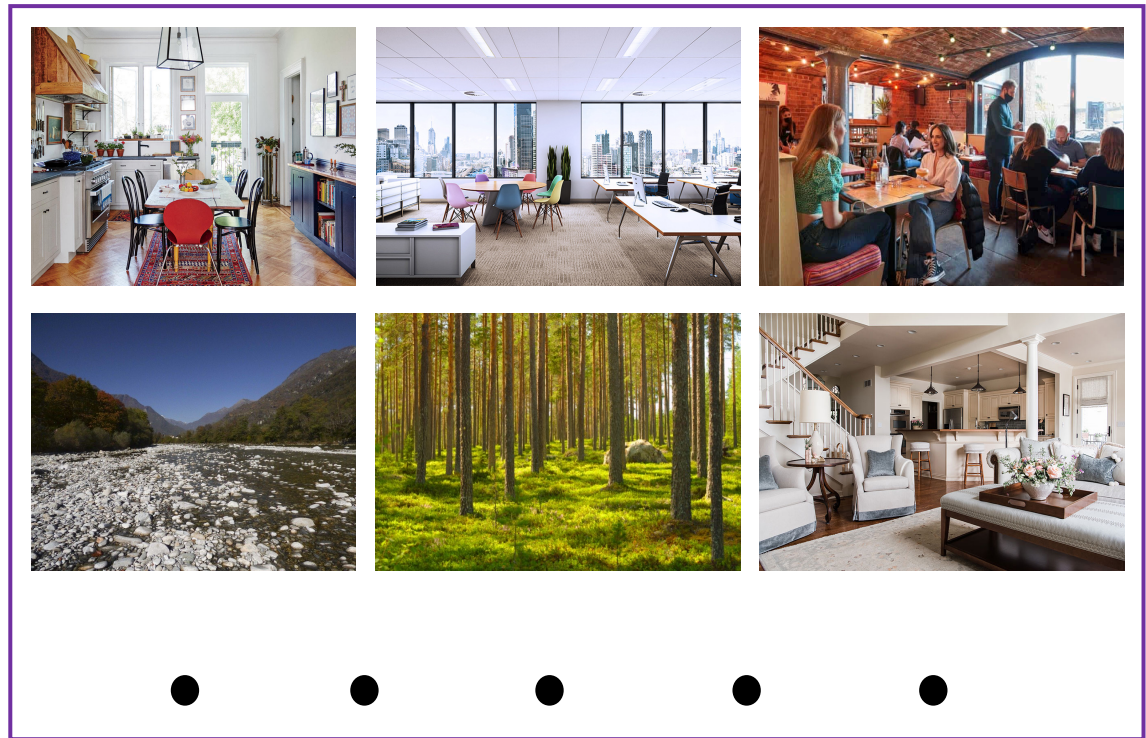
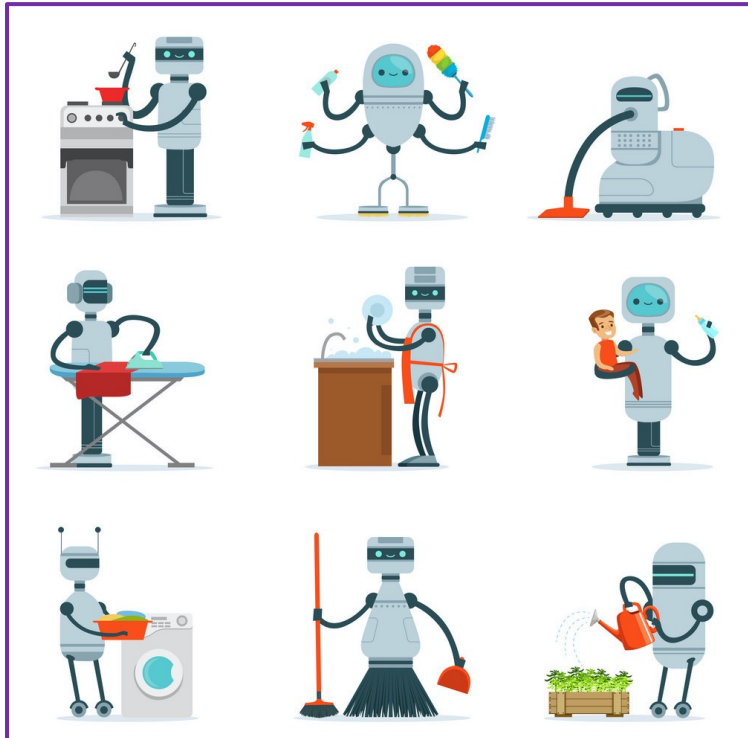
- ❑ Ultimate goal: Build **general-purpose embodied** intelligence by learning to make sequential decisions in the physical world



What is the Goal of “Robot Learning”?

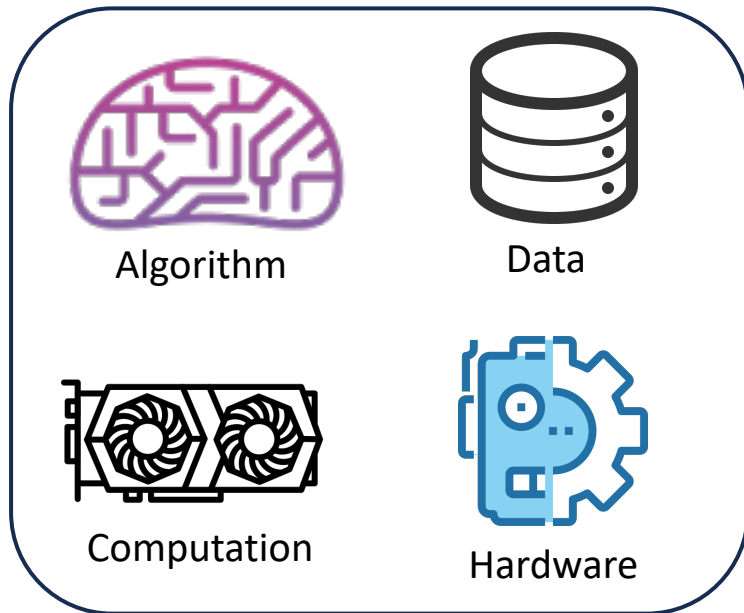
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Robots that can do thousands of tasks in thousands of environments



What is the Goal of “Robot Learning”?

- ❑ Ultimate goal: Build **general-purpose embodied** intelligence by learning to make sequential decisions in the physical world



How far are we?



Embodied AGI in the real physical world



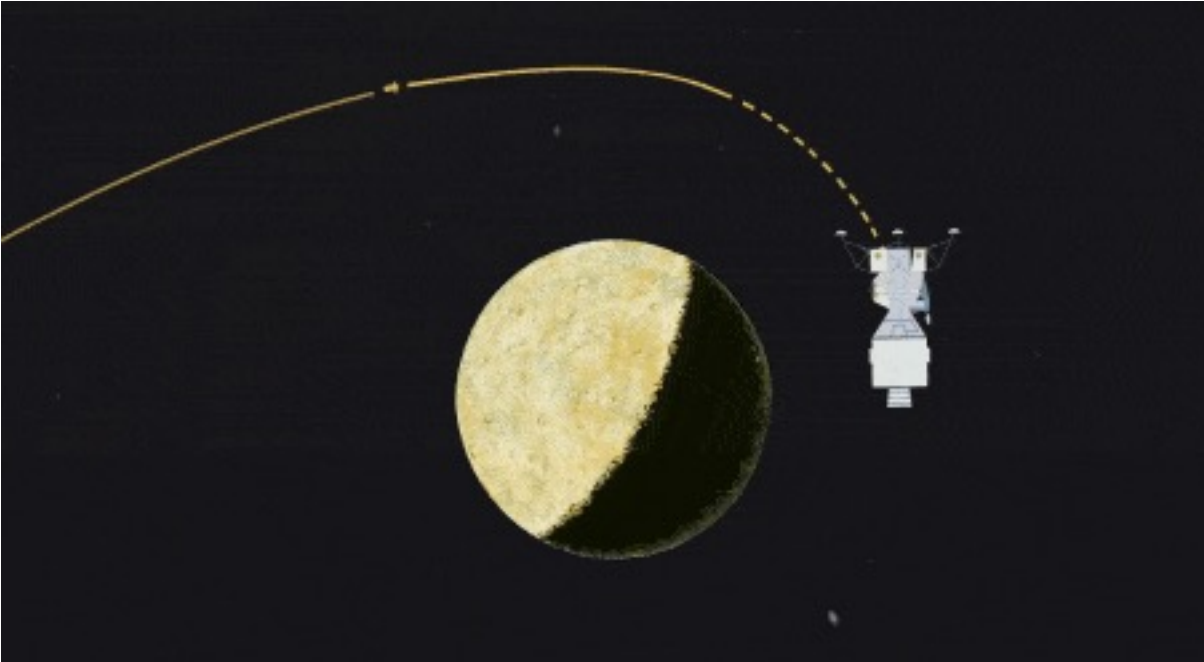
- ❑ We’ve made (a lot of) progress in terms of **domain-specific embodied intelligence**, either learning-based or not
- ❑ However, we are still very far from general-purpose embodied intelligence!

Lecture 1&2's Agenda

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Where are We Today: Non-learning Method

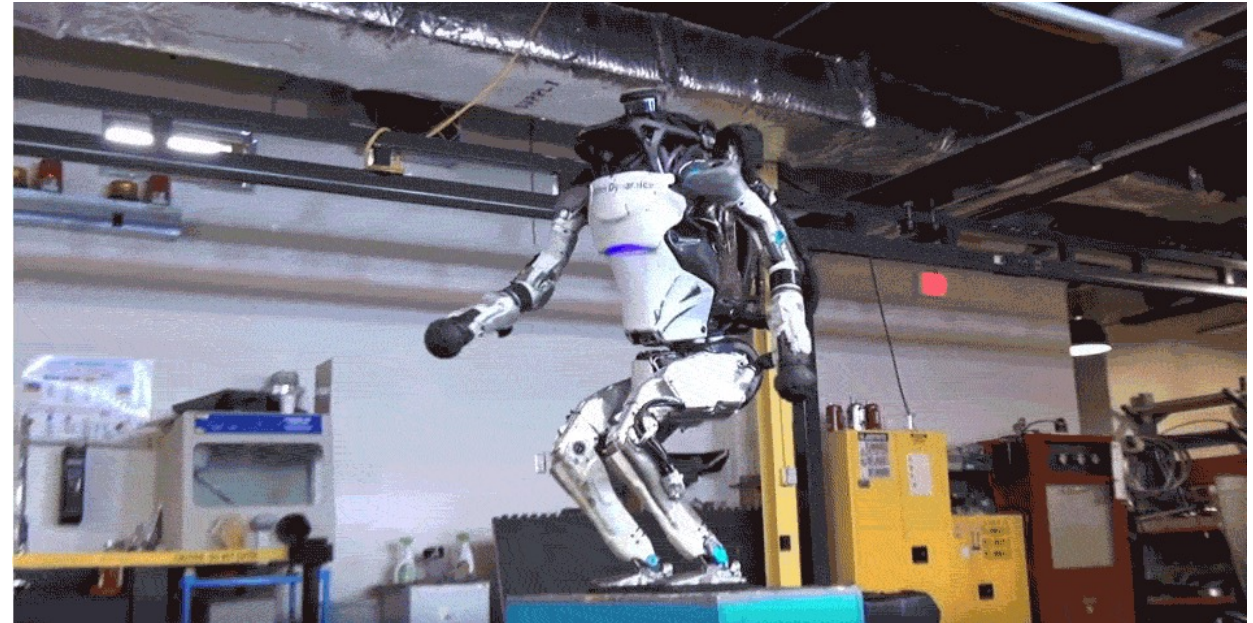
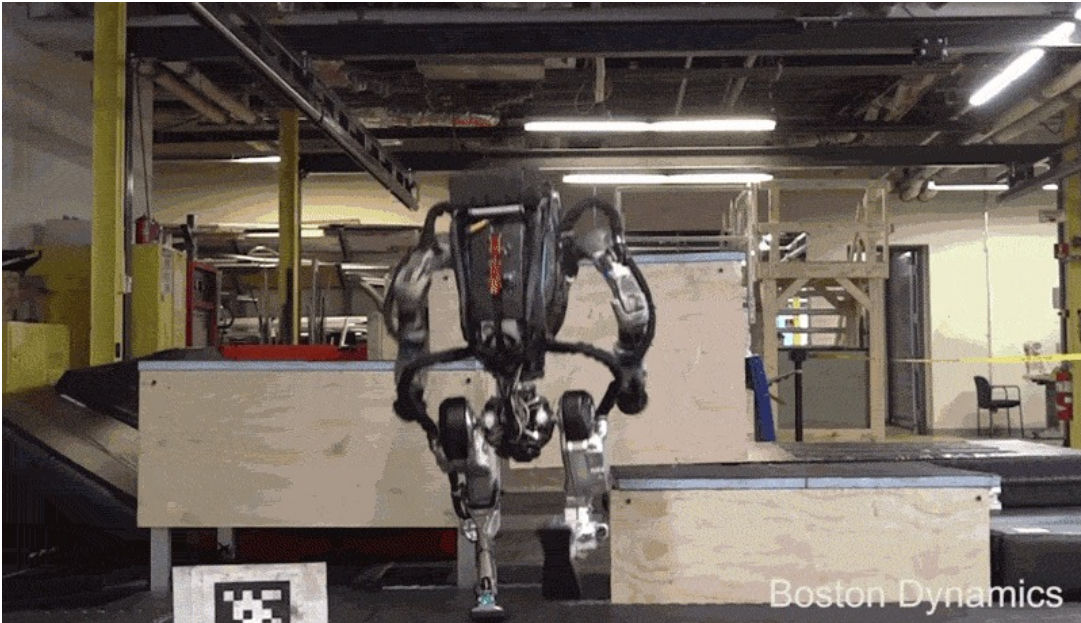
- ❑ Before seeing today's progress in robot learning, let's first see some results w/o learning:



Apollo 11 trajectory optimization and control: optimal control + robust control

Where are We Today: Non-learning Method

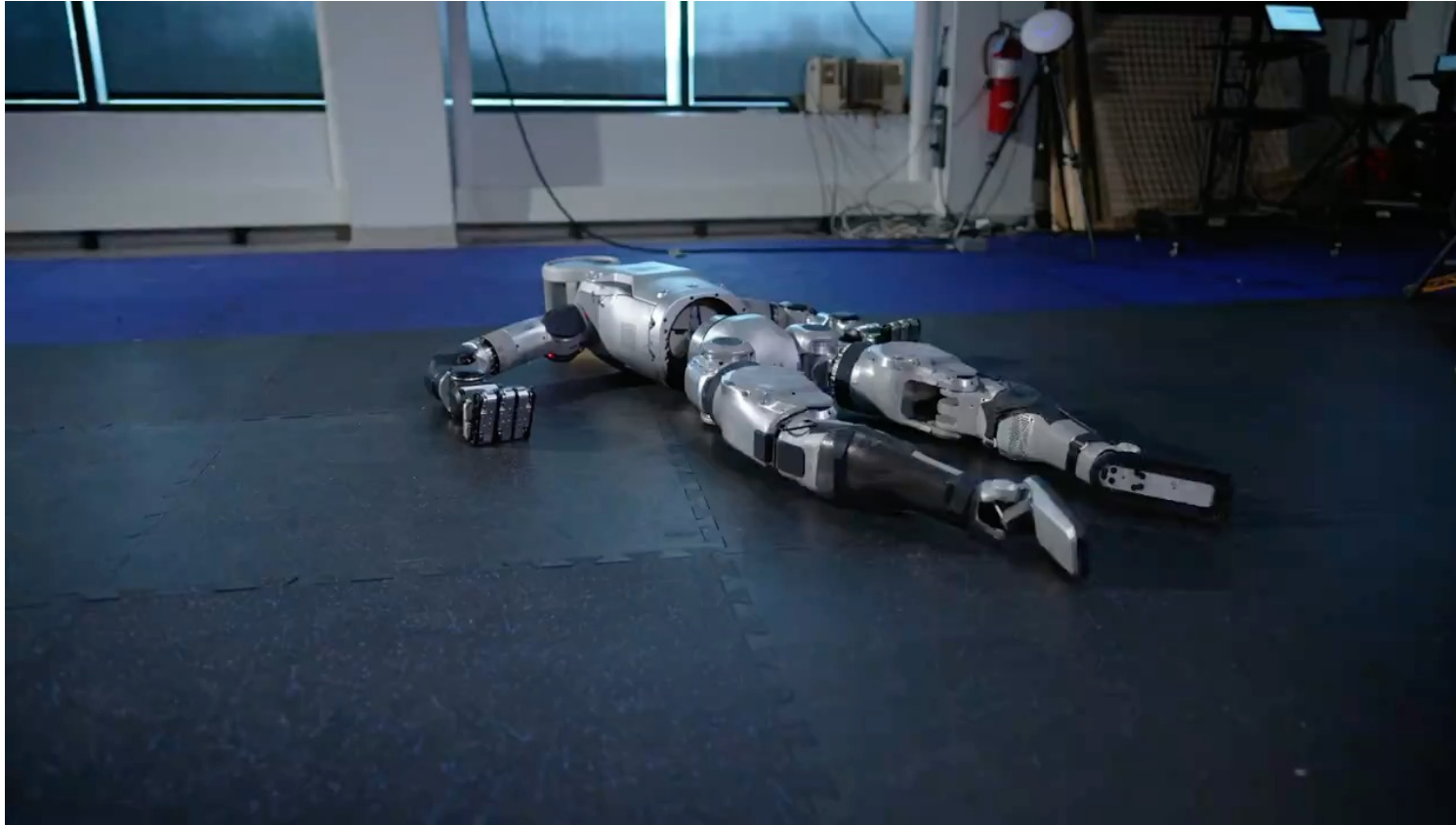
- ❑ W/o learning:



Boston Dynamics Atlas humanoid planning and control: trajectory optimization + MPC

Where are We Today: Non-learning Method

- ❑ The legendary Atlas just retired on April 16, 2024



New Atlas (100% electric motors)

Where are We Today: Non-learning Method

❑ W/o learning:

Model Predictive Control for Aggressive Driving Over Uneven Terrain

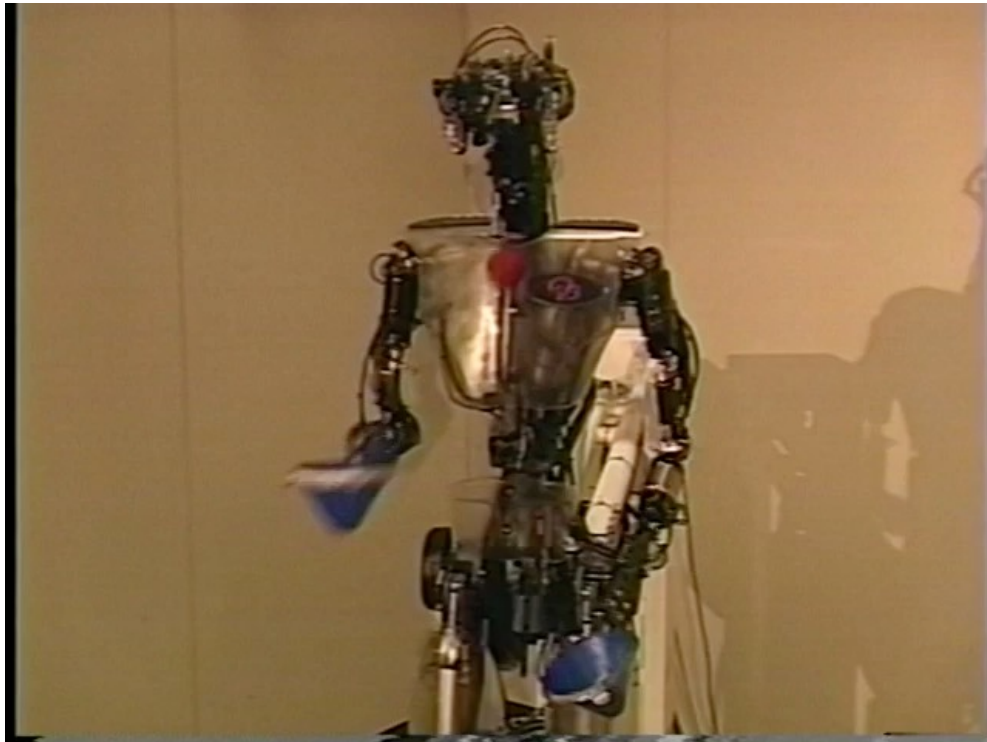
Tyler Han*, Alex Liu, Anqi Li, Alexander Spitzer, Guanya Shi, and Byron Boots



Offroad autonomy: sampling-based MPC

Where are We Today: Non-learning Method

❑ W/o learning:



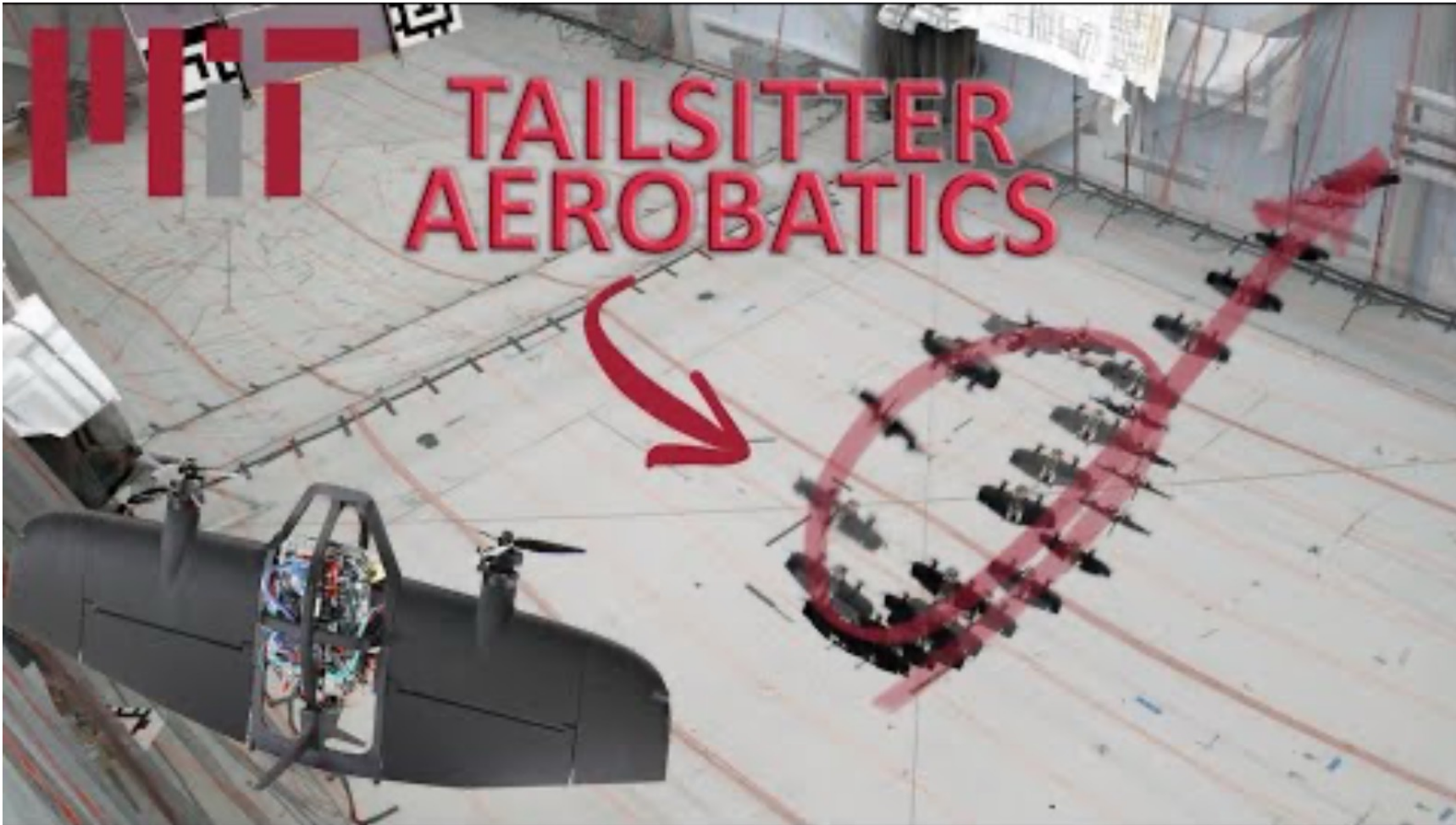
(1990s) Humanoid juggling
Schaal and Atkeson



High-speed robot hand (Ishikawa Komuro Lab) in 2009

Where are We Today: Non-learning Method

- ❑ W/o learning:



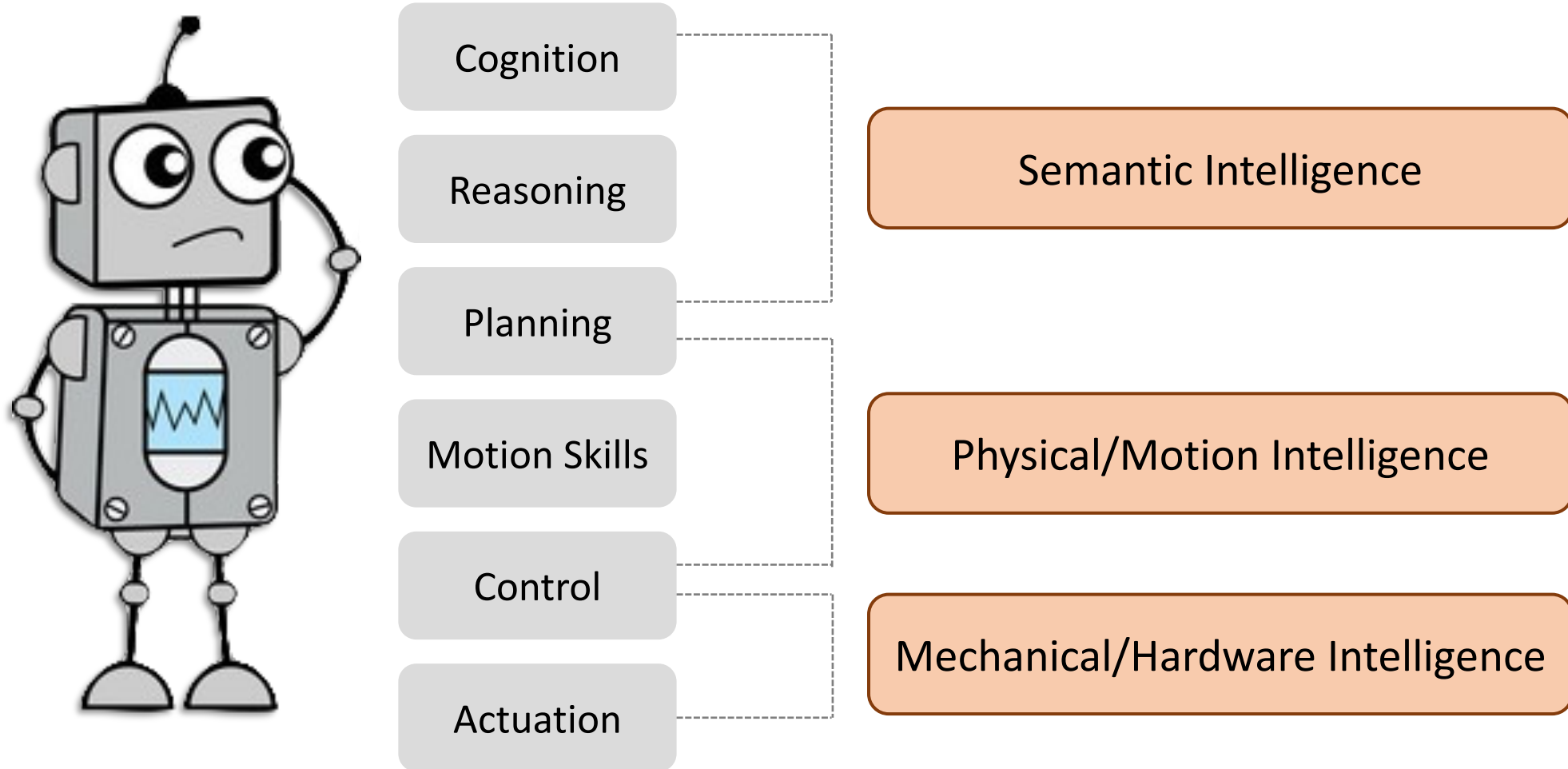
Tailsitter aerobatics (MIT LIDS): trajectory generation + nonlinear control

Why Do We Need Robot Learning?

- ❑ Data increases the “upper bound” or reduces/relax certain limitations of algorithm, computation, and hardware
 - Modeling the world could be hard
 - Environments/tasks might change
 - The policy structure might be limited
 - The optimization solver might be suboptimal
 - Assumptions might not hold

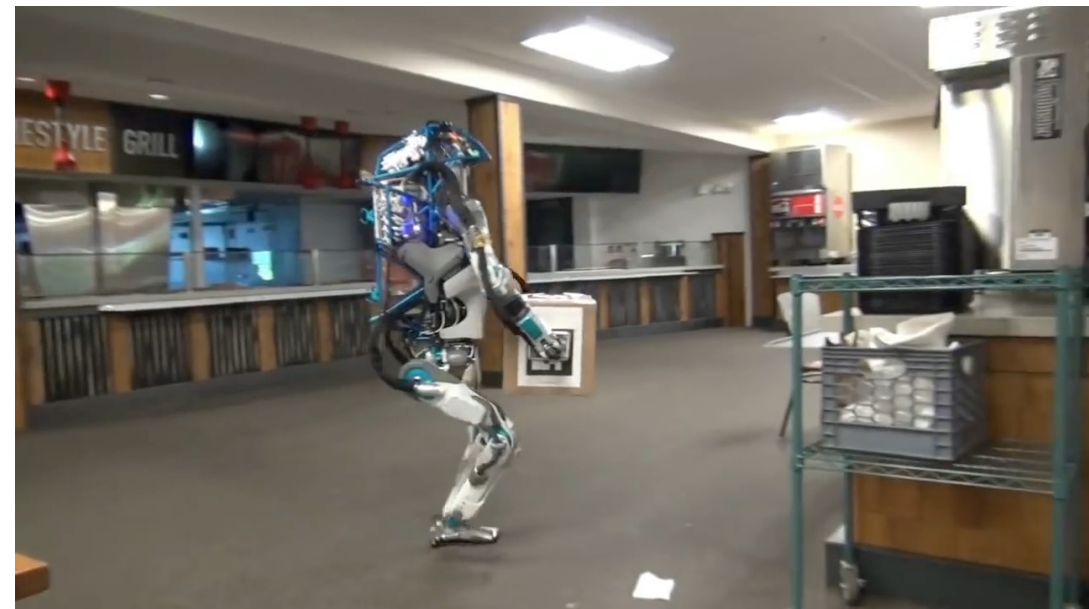
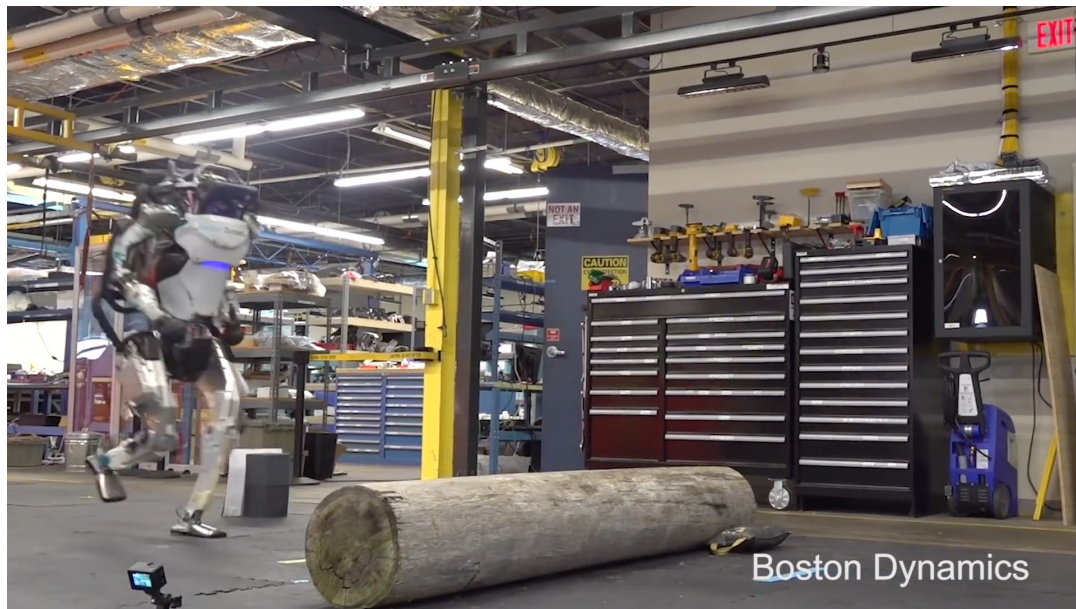
Why Do We Need Robot Learning?

- ❑ Non-learning methods often separately design each module in robotics



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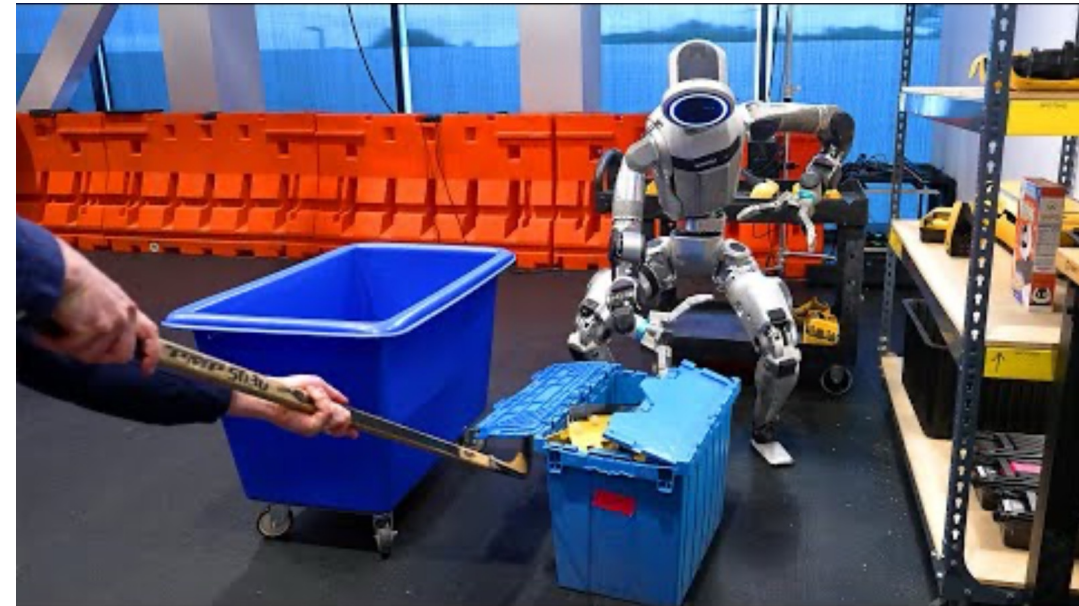


Why Do We Need Robot Learning?

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Robot learning!

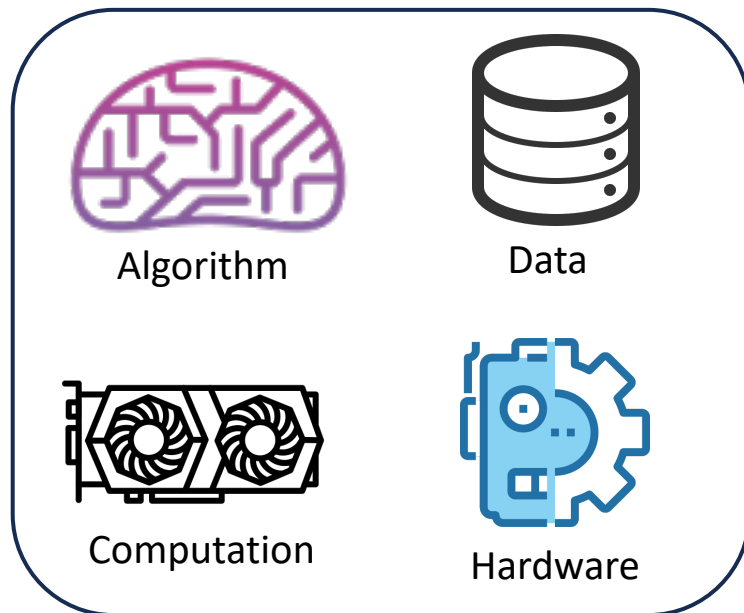


Atlas + Large Behavior Models

<https://bostondynamics.com/blog/large-behavior-models-atlas-find-new-footing/>

Next Time: A Glimpse of Today's Robot Learning

- ❑ In this class, we will learn how these methods work and more!
 - Model-free reinforcement learning (RL)
 - Sim2Real and Real2Sim2Real
 - Online adaptation
 - Imitation learning from demonstrations
 - Model-based RL
 - Offline RL



How far are we?



Embodied AGI in the real physical world



Thank You!